Supply Chain Black Box: Hidden Supplier Dependency Discovery, Calibrated Cascading-Risk Prediction and Optimised Recovery for an FMCG Manufacturer

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***Abstract*—A chocolate and malt-beverage manufacturer sees its Tier-1 suppliers but not the Tier-2/3 suppliers they share, so a single hidden trader can starve several "independent" cocoa processors at once. We build, evaluate and deploy an end-to-end decision system that (i) predicts 30-day supplier disruption with calibrated probabilities from point-in-time features, (ii) ranks candidate hidden relationships by link prediction, (iii) propagates disruptions through a daily material-flow simulator and (iv) converts predictions into cost-minimising actions and a mixed-integer recovery plan. Supplier-level experiments use a documented synthetic FMCG ecosystem (500 suppliers, 1,500 links, 69 % hidden); two real public datasets test transferable components (DataCo, 172,765 shipments; SupplyGraph, a real FMCG graph). Platt scaling halves XGBoost's calibration error (0.020→0.010); the observed network understates Tier-2 and Tier-3 exposure 2.1× and 10.8×; graph neural networks did *not* beat feature-based models. In a predefined nine-scenario evaluation (500 common-random-number replications) optimisation-based recovery cuts the simulated impact of a severe cocoa-trader outage from INR 4.72 to 1.16 crore (75 %), saving 3.56 crore (95 % CI 3.25–3.87), while over-hedging under mild shocks.**

***Index Terms*—supply chain risk, link prediction, graph neural networks, probability calibration, cascading disruption, recovery optimisation, FMCG**

I. Introduction

A. Business Problem and Relevance

Announced supply chain disruptions are followed by significantly lower long-run stock returns and higher equity risk [1]. In FMCG the exposure is structural: a malt-beverage jar needs malt extract, milk powder, cocoa, sugar, vitamin premix and PET jars, and each Tier-1 input depends on Tier-2 commodity traders that procurement rarely maps, so disruptions "ripple" through invisible links [2]. We support two executable decisions: *(1) each month, which mitigation action should procurement take for each supplier; (2) when a shock is reported, which alternative suppliers should receive which quantities, by which transport mode?*

B. Research Questions and Contributions

RQ1 Can point-in-time features yield *calibrated* 30-day disruption probabilities? **RQ2** How much exposure is hidden in unobserved links, and can link prediction recover them? **RQ3** Does graph structure improve risk classification over tabular models? **RQ4** Does network-aware optimisation outperform rule-based recovery across disruption severities? Contributions: (1) an integrated, deployed pipeline in which every recommendation is traceable to a model output, an exposure calculation and a solver result; (2) a leakage-audited protocol (prediction timestamp, embargoed chronological splits, calibration-only split, test edges removed from message passing); (3) a quantification of the *black-box gap*; (4) a predefined nine-scenario policy evaluation with explicit economics and paired statistics. We do not claim novelty for GNNs, calibration or MILP individually.

II. Related Work

Risk prediction. AI for supply chain risk management concentrates on identification and assessment, with few studies closing the loop to response [3]. Gradient boosting [4] is a strong tabular baseline; because actions are triggered by probabilities, calibration matters: Platt scaling [5], isotonic regression [6] and their comparison across learners [7]. **Graph intelligence.** Brintrup *et al.* [8] framed hidden-dependency discovery as link prediction on a real automotive network; Kosasih and Brintrup [9] applied GNNs to the same task. Heuristics [10], node2vec [11], GCN [12], GraphSAGE [13] and GAT [14] are our comparators; SupplyGraph [15] is a real FMCG benchmark. **Cascades and recovery.** OR/MS research models disruption mitigation through inventory, sourcing and contracts [16]; the ripple-effect literature studies propagation [2]. Table I shows that prior work addresses these stages separately; our contribution is their leakage-controlled integration with a measured business outcome.

III. Methodology

A. Data and Provenance

Synthetic FMCG ecosystem (fictional "Cocoa & Malt Foods", seed 42): 120/200/180 Tier-1/2/3 suppliers, 50 materials over three BOM levels, 100 products, two plants, 1,048 disruptions (782 qualifying) in 2024–2025, 4,929 POs, 9,579 shipments and 5,824 weekly inventory records. A buyer

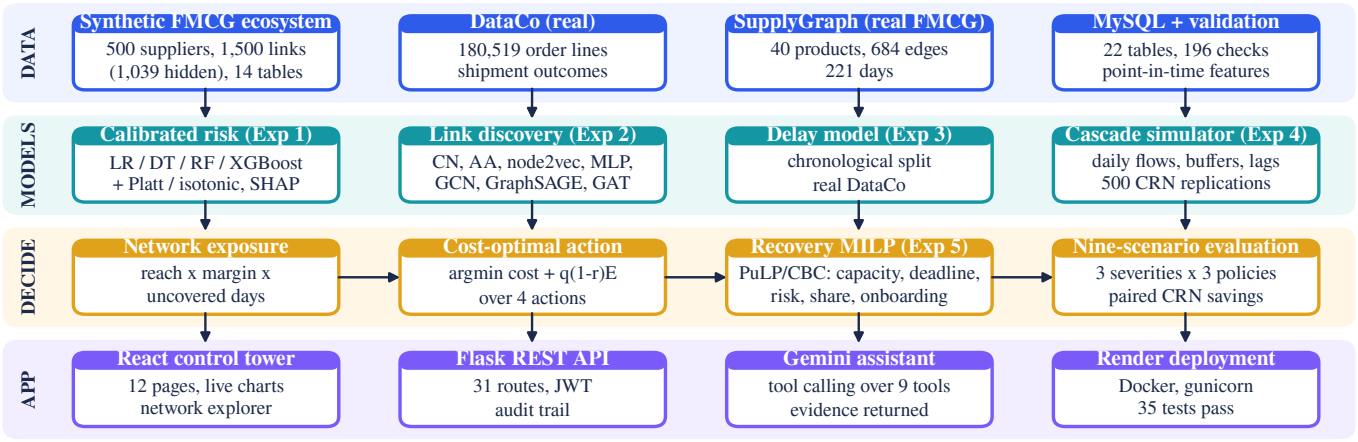


Fig. 1. System architecture. Real and synthetic data feed point-in-time features; calibrated risk, link discovery, delay and cascade models feed an exposure model, a cost-optimal action rule and a recovery MILP; results are served by a Flask API, a React control tower and a tool-calling Gemini assistant, deployed on Render.

TABLE I
SCOPE OF RELATED WORK (✓ PRIMARY FOCUS, ○ PARTIAL / REVIEW).

Work	Hidden links	Risk pred.	Calib- ration	Cas- cade	Recovery opt.	Policy eval.
Brintrup *et al.* [8]	✓					
Kosasih & Brintrup [9]	✓					
Wasi *et al.* [15]		✓				
Baryannis *et al.* [3]		○			○	
Snyder *et al.* [16]				○	✓	○
Ivanov *et al.* [2]		○		✓		
This work	✓	✓	✓	✓	✓	✓

needing material m picks $k \in \{2..6\}$ sources with probability $\propto \text{size}^{1.6} e^{-d/2500\,\text{km}}$; Tier-2 and Tier-3 links are visible with probability 0.45 and 0.15 (461 of 1,500 observed). The daily disruption hazard combines latent financial risk, utilisation, regional geo-risk, monsoon/cyclone seasonality, an unobserved frailty and contagion from disrupted upstream suppliers over the *true* network; hidden links are used only as simulator ground truth and for evaluation. **DataCo** [17] (CC BY 4.0, SHA-256 recorded): 180,519 order lines, 7,754 cancelled removed. **SupplyGraph** [15]: 40 products (one duplicated node removed), 684 edges, 221 days. Real and synthetic identities are never merged; all tables pass 196 validation checks (keys, foreign keys, dates, ranges, logical rules).

B. Disruption Prediction and Leakage Control (RQ1)

The prediction timestamp t is 00:00 on the first of each month. For supplier i,

$$y_i(t) = \mathbb{1}\big[\exists e:\ s_e \in (t, t+30\text{d}],\ \ell_e \geq 0.25,\ d_e \geq 3\big], \quad (1)$$

with start s_e, capacity loss ℓ_e and duration d_e. The 25 features use only disruptions reported before t, deliveries completed before t, the monthly risk forecast and *observed* links. Splits: train 2024-04..2025-02 (5,500 rows), one-month embargo, calibration 2025-04..06 (1,500), test 2025-07..12 (3,000). Platt scaling $q = \sigma(a\,\text{logit}\,\hat{p} + b)$ and isotonic regression are fitted on the calibration split only and chosen by 5-fold CV Brier score; we report Brier and ECE $= \sum_b \frac{n_b}{N} |\bar{y}_b - \bar{q}_b|$ (10 equal-width bins).

C. Graph Learning (RQ2, RQ3)

$G = (V, E)$ has 501 nodes with 131 observable attributes. GraphSAGE updates $h_v^{(l+1)} = \text{ReLU}(W_1 h_v^{(l)} + W_2\,\text{mean}_{u \in N(v)} h_u^{(l)})$ and a decoder scores $\sigma(\text{MLP}[z_u \odot z_v, |z_u - z_v|])$. **A1**: observed links split 70/10/20 with test links removed from message passing; negatives are random or *material-compatible*. **A2**: train on all observed links and rank all 8,863 compatible unobserved pairs against 1,039 hidden links; the downstream model is chosen by validation AP only. **Exp. 2B** feeds Exp. 1 features through GCN/GraphSAGE/GAT on observed, observed+top-500 predicted and oracle graphs, and predicts a 7-day service shortfall on SupplyGraph.

D. Exposure, Cascade Simulation and Decision Rule

Supplier i's reach r_{im} multiplies dependency shares along paths to the manufacturer; concentration is $\text{HHI}_m = \sum_s w_{sm}^2$ and

$$E_i = \sum_m r_{im}\,\text{CM}_m\,\mathbb{E}\big[S\max(0, D - c_m - 10(\tau_i - 1))\big], \quad (2)$$

with daily contribution margin CM_m, cover c_m, tier τ_i and empirical severity/duration (S, D). The simulator propagates availability daily over the true network, $a_b(t) = \min\big(1 - \ell_b(t), \min_i \min(1, A_{bi}(t))\big)$ with $A_{bi}(t) = I_{bi}(t) + \sum_u w_{ub} a_u(t - L_{ub})$, 10-day buffers, lead times L_{ub} and BOM-bottleneck production. The executable action is

$$a^* = \arg\min_a\ C_a + q_i(1 - \rho_a)E_i, \quad (3)$$

over four actions (C_a=0/5k/20k/60k INR, ρ_a=0/0.15/0.45/0.70), Bayes-optimal under calibration with thresholds $q^* = \Delta C/(E_i \Delta\rho)$.

TABLE II

EXP. 1: 30-DAY SUPPLIER DISRUPTION PREDICTION ON THE UNTOUCHED TEST SPLIT (3,000 SUPPLIER-MONTHS, PREVALENCE 7.4%). CALIBRATORS FITTED ON THE CALIBRATION SPLIT ONLY; SHADED = DEPLOYED.

Model	Calib.	ROC-AUC	PR-AUC	Brier	ECE_{10}
Logistic reg.	–	0.666	0.150	0.0669	0.0066
Logistic reg.	Platt	0.666	0.150	0.0670	0.0059
Logistic reg.	Iso.	0.659	0.126	0.0687	0.0070
Decision Tree	–	0.571	0.098	0.0704	0.0366
Decision Tree	Platt	0.571	0.098	0.0689	0.0111
Decision Tree	Iso.	0.569	0.094	0.0689	0.0124
Random Forest	–	0.657	0.140	0.0672	0.0046
Random Forest	Platt	0.657	0.140	0.0675	0.0081
Random Forest	Iso.	0.655	0.124	0.0673	0.0063
XGBoost	–	0.626	0.122	0.0689	0.0197
XGBoost	Platt	0.626	0.122	0.0680	0.0104
XGBoost	Iso.	0.620	0.113	0.0681	0.0109

E. Recovery Optimisation and Nine-Scenario Design (RQ4)

For deficits δ_m the MILP (PuLP/CBC) chooses quantities x_{jk} from candidate j by mode k, onboarding z_j and shortage s_m:

$$\min \ \sum_{jk} p_j(1+f_{jk}+hW/2)x_{jk} + \sum_j F z_j + \sum_m \pi_m s_m \tag{4}$$

$$\text{s.t. } \sum_{jk} \rho_j(1-q_j)\kappa_j\alpha_j x_{jk} + s_m \geq \delta_m,$$
$$x_{jk} \leq u_j(W-L_{jk})^+, \ \ \sum_k x_{jk} \leq 0.6\,\delta_m,$$
$$x_{jk} = 0 \text{ if } q_j > 0.35, \ \ \sum_k x_{jk} \leq M z_j,$$

with compatibility κ_j, spare capacity u_j, deadline W, freight f_{jk}, holding rate h, penalty π_m and availability α_j derated by estimated dependency on the disrupted suppliers. The shock hits the five largest Tier-2 cocoa-bean traders (19 observed, 25 hidden links) with 25/50/100 % capacity loss for a nominal 60 days. Policies: **A** none; **B** rule-based (observed graph, same-material alternates by reliability, reactive at 10 days of cover); **C** optimisation (observed+predicted graph, MILP at detection plus the same trigger). All policies share network, demand, inventory and economics, and replication r uses seed $s_0 + r$ for every policy (common random numbers, 180 days). Illustrative economics (INR): premium 6 % (10 % substitutes), onboarding 1.5 L, freight 2.5/4 % (expedited 3×), holding 24 %/yr, shortage penalty 15 % of price, interruption 75k per plant-day, recovery management 25k/day.

IV. RESULTS

A. Calibrated Disruption Prediction (RQ1)

Discrimination is modest, as expected for rare, partly contagion-driven events (prevalence 7.4 %; Table II, Fig. 2): logistic regression ranks best (ROC-AUC 0.666, PR-AUC 0.150, twice the base rate) and the deployed XGBoost reaches 0.626/0.122. Calibration is decisive for decisions: Platt reduces ECE from 0.020 to 0.010 (XGBoost) and from 0.037 to 0.011 (tree). On the test months the cost-based policy raised 5 contingency alerts; realised cost was 118.3 lakh vs. 116.9 (monitor only), 120.4 (uncalibrated) and 664.3 (review all), with a perfect-foresight bound of 82.1. Because 3–4 weeks of inventory absorb most single-supplier events, value shifts to network-level response—a finding that motivates RQ2–RQ4.

TABLE III

EXP. 2A LINK PREDICTION (SYNTHETIC, 5 SEEDS FOR LEARNED MODELS). A1: HELD-OUT OBSERVED EDGES VS. MATERIAL-COMPATIBLE NEGATIVES. A2: RANKING ALL 8,863 COMPATIBLE UNOBSERVED PAIRS AGAINST 1,039 GROUND-TRUTH HIDDEN EDGES (BASE RATE 0.117).

	A1 (held-out)		A2 (hidden-edge discovery)		
Method	ROC-AUC	AP	AP	P@100	R@500
Common neigh.	0.500	0.500	0.117	0.06	0.042
Adamic-Adar	0.500	0.500	0.117	0.06	0.042
Pref. attach.	0.733	0.763	0.166	0.34	0.120
Node2Vec	0.638	0.609	0.146	0.24	0.084
MLP	0.826±0.017	0.849±0.012	0.227	0.32	0.141
GCN	0.807±0.020	0.837±0.010	0.176	0.25	0.109
GraphSAGE	0.737±0.009	0.784±0.012	0.168	0.29	0.102
GAT	0.796±0.005	0.811±0.018	0.177	0.26	0.103
GraphSAGE+skip	0.792±0.014	0.827±0.006	0.166	0.29	0.102

TABLE IV

EXP. 2B: GRAPH VS. TABULAR RISK CLASSIFICATION (TEST SPLIT, PLATT-CALIBRATED; MEAN±SD OVER 5 SEEDS). SYNTHETIC: 30-DAY DISRUPTION (PREVALENCE 7.4%); SUPPLYGRAPH (REAL FMCG): 7-DAY SERVICE SHORTFALL (PREVALENCE 28.8%).

Data	Model [graph]	PR-AUC	ROC-AUC	Brier
Synthetic	XGBoost (tabular)	0.122±0.000	0.626	0.0680
Synthetic	MLP (no graph)	0.104±0.011	0.580	0.0685
Synthetic	GCN [observed]	0.104±0.005	0.590	0.0684
Synthetic	GCN [obs.+pred.]	0.111±0.011	0.591	0.0683
Synthetic	GCN [oracle]	0.107±0.004	0.594	0.0684
Synthetic	GraphSAGE [observed]	0.105±0.004	0.588	0.0685
Synthetic	GraphSAGE [obs.+pred.]	0.106±0.007	0.579	0.0685
Synthetic	GraphSAGE [oracle]	0.099±0.006	0.587	0.0685
Synthetic	GAT [observed]	0.100±0.009	0.585	0.0686
Synthetic	GAT [obs.+pred.]	0.098±0.010	0.585	0.0686
Synthetic	GAT [oracle]	0.108±0.005	0.598	0.0683
SupplyGraph	Logistic Regression	0.488±0.000	0.701	0.1841
SupplyGraph	XGBoost (tabular)	0.454±0.000	0.668	0.1905
SupplyGraph	MLP (no graph)	0.475±0.007	0.701	0.1905
SupplyGraph	GCN	0.459±0.030	0.633	0.1961
SupplyGraph	GraphSAGE	0.461±0.017	0.607	0.1975
SupplyGraph	GAT	0.443±0.040	0.660	0.1932

B. The Black-Box Gap and Hidden-Link Discovery (RQ2, RQ3)

Using only observed links understates Tier-2 and Tier-3 exposure 2.1× and 10.8× (Fig. 3). For discovery (Table III), common neighbours and Adamic–Adar are uninformative on the layered network (no shared neighbours); the validation-selected MLP pair model attains AP 0.227 against a base rate of 0.117 (P@100 = 0.32), whereas GraphSAGE reaches 0.168. Graph structure did not improve risk classification (Table IV)—not even with the oracle graph—and on real SupplyGraph logistic regression is best. Predicted links are therefore treated as *verification leads*, never as confirmed relationships.

C. Transfer to Real Shipment Data (DataCo)

On 34,554 chronologically held-out order lines XGBoost reaches ROC-AUC 0.780 (PR-AUC 0.858) vs. 0.744 for a shipping-mode baseline (Table V); delay-days regression adds nothing over the mode mean (R^2 0.27 vs. 0.28).

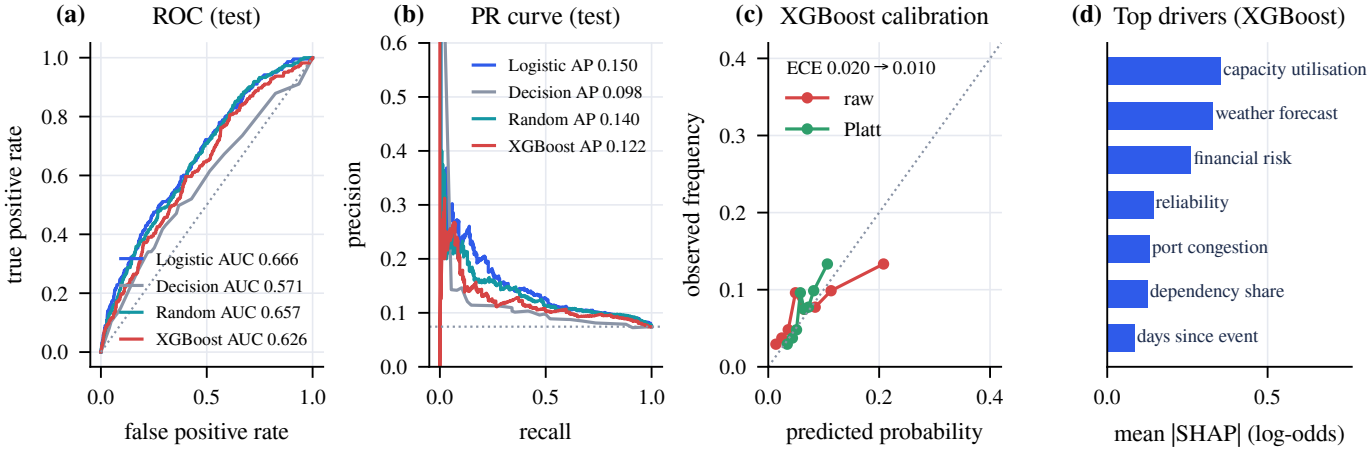


Fig. 2. Exp. 1 on the untouched test split (3,000 supplier-months): (a) ROC and (b) precision–recall of the four calibrated models, (c) XGBoost reliability before/after Platt scaling fitted on the calibration split, (d) global SHAP attributions.

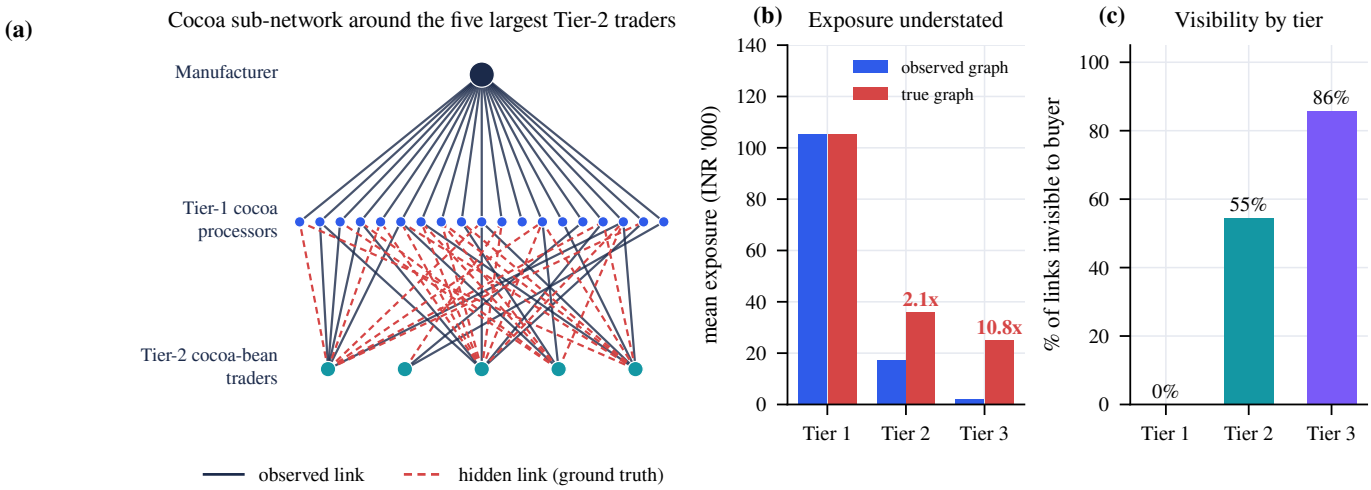


Fig. 3. The black-box gap (synthetic ground truth): (a) cocoa sub-network around the five largest Tier-2 traders—dashed links are invisible to the buyer; (b) mean exposure on the observed vs. true network; (c) share of links invisible by supplier tier.

TABLE V
EXP. 3 (REAL DATACO): LATE-DELIVERY PREDICTION, CHRONOLOGICAL TEST SPLIT (34,554 ORDER LINES, LATE RATE 57.3%), THRESHOLD 0.5.

Model	ROC-AUC	PR-AUC	F1	Brier
Logistic Regression	0.749	0.839	0.682	0.1878
Random Forest	0.777	0.855	0.710	0.1768
XGBoost	0.780	0.858	0.713	0.1768
Shipping-mode rate	0.744	0.785	0.670	0.1881

TABLE VI
WORKED DECISION FOR SUPPLIER S0065 (TIER-1, SYNTHETIC) AT t=2025-12-01, PRODUCED END-TO-END BY THE DECISION ENGINE.

Raw / calibrated 30-day risk: 40.8 % / **15.2 %** Exposure E_i: INR 4.41 lakh
Break-even q^*: 7.6 % / 11.3 % / 36.3 % Cover: 24 d
SHAP drivers: capacity utilisation (+0.67), mean delay (90 d) (+0.41), dependency share (+0.30)

Action	Expected cost (INR '000)
Routine monitoring	66.9
Enhanced monitoring	61.9
Contingency review	56.8 ←
Escalate & recover	80.1
Recovery MILP: Optimal, cost INR 15.1k	

D. Executable Decision Example

Table VI traces one recommendation end to end: the calibrated probability, network exposure and break-even thresholds select a contingency review as the minimum expected-cost action, and the recovery MILP returns an optimal standby allocation. Every value is reproduced by `POST /api/decision/evaluate` and stored in the audit trail.

E. Nine-Scenario Business Evaluation (RQ4)

Table VII and Fig. 4 report all thirteen outcomes. Mild shocks are absorbed by buffers for every policy. Under the moderate shock impacts are 61.5 (A), 50.5 (B) and 44.8 lakh

TABLE VII

NINE-SCENARIO EVALUATION (MEANS OF 500 COMMON-RANDOM-NUMBER REPLICATIONS; SIMULATED INR ON SYNTHETIC DATA). SEVERITY = CAPACITY LOSS OF THE FIVE LARGEST TIER-2 COCOA-BEAN TRADERS FOR A NOMINAL 60 DAYS (LOG-NORMAL). POLICY A NONE, B RULE-BASED, C OPTIMISATION. DISRUPTED CAPACITY (TIME-AVERAGED OVER THE HORIZON) IS IDENTICAL ACROSS POLICIES WITHIN A SEVERITY.

Scen.	Sev.	Pol.	Aff. sup.	Aff. mat.	Unmet (k units)	Inv. short. (L)	Procure. (L)	Transport (L)	Impact, Cr [95% PI]	Recov. d	Alt. util. %	Full rec. %
S1	Mil.	A	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	0	100
S2	Mil.	B	10.4	2.4	0.0	0.0	0.0	0.00	0.26 [0.00, 0.45]	101	1	100
S3	Mil.	C	10.4	2.4	0.0	0.0	4.8	0.12	0.26 [0.01, 0.46]	101	5	100
S4	Mod.	A	17.1	3.9	47.6	2.8	0.0	0.00	0.61 [0.23, 4.54]	123	0	82
S5	Mod.	B	17.1	3.9	29.8	1.9	3.0	0.12	0.51 [0.23, 3.70]	123	16	88
S6	Mod.	C	17.1	3.7	2.6	0.2	86.2	2.95	0.45 [0.38, 0.63]	123	16	97
S7	Sev.	A	19.8	4.0	734.1	46.3	0.0	0.00	4.72 [0.23, 16.10]	127	0	14
S8	Sev.	B	19.8	4.0	605.3	38.9	18.3	0.73	3.98 [0.24, 15.19]	126	42	15
S9	Sev.	C	19.8	3.7	39.2	5.3	461.9	24.20	1.16 [0.93, 3.84]	125	23	88

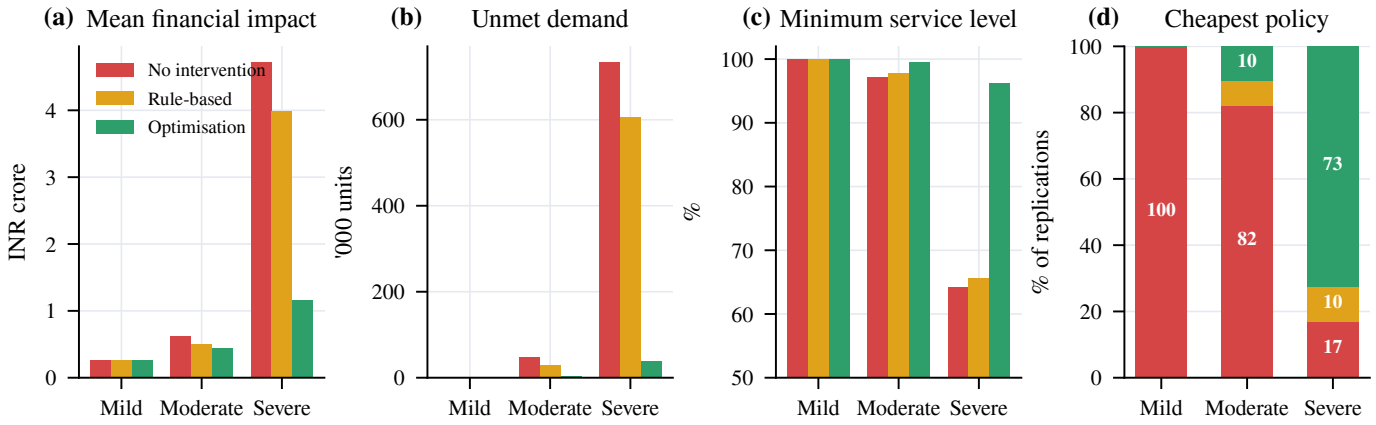


Fig. 4. Nine-scenario outcomes over 500 common-random-number replications (simulated INR on synthetic data): (a) mean financial impact, (b) unmet demand, (c) minimum service level, (d) share of replications in which each policy is cheapest.

(C). Under the severe shock, no intervention leaves 734k units unmet (minimum service 64 %, 4.72 crore); the rule-based policy reaches 605k (3.98 crore) because it can exclude only processors with *observed* links to the failed traders and reacts after cover falls. Optimisation pre-books de-rated alternates and expedited freight, cutting unmet demand to 39k units with full recovery in 88 % of replications (1.16 crore). Paired CRN savings of C are 3.56 crore over A (95 % CI 3.25–3.87) and 2.82 crore over B; C is cheapest in 73 % of severe replications, whereas no intervention is cheapest in 82 % of moderate ones and C over-hedges by 0.82 lakh under mild shocks. The relative standard error of the severe mean is 3.9 % and recovery time is censored in at most 3 % of replications.

V. DISCUSSION

Relevance. The system answers a concrete FMCG procurement question with an auditable action and allocation (Table VI, Fig. 5). **Significance.** Value concentrates in rare, severe network shocks: a 75 % reduction in simulated impact, with explicit economics and confidence intervals, supporting a tiered policy—monitor routinely, optimise when the reported shock is large. **Originality.** The contribution is the leakage-controlled integration and the measurement of what hidden links cost (Table I); the negative GNN findings, including the oracle-graph ablation, are informative for firms with sparse supplier maps. **Technical quality and validity.** Labels follow a fixed timestamp; splits are chronological with an embargo; calibrators never see test labels; link models are selected on validation data; policies share common random numbers; 61 automated checks recompute every reported metric. **Limitations.** Supplier-level results rest on synthetic data whose generator we designed; real-network performance is not established; economics are illustrative; predicted-link quality is moderate; the static MILP over-hedges for mild shocks; Gemini's free tier is rate-limited, so a deterministic tool router answers when it is busy.

VI. REPRODUCIBILITY

Code, data, configurations and logged runs: https://github.com/SamarthKamat/AIML-PROJECT (branch `claude/exciting-euler-5v1r2l`); live demo: https://supply-chain-black-box.onrender.com. `bash run_all.sh` downloads DataCo and SupplyGraph at pinned commits with hash checks, regenerates data (seed 42), runs Exps. 1–5 (GNN seeds 0–4), runs 61 reconciliation checks (confusion matrices, split counts, prevalence, recomputed metrics, CRN integrity, impact components), rebuilds every table, figure and quoted number of this paper from result files, and runs 35 tests on SQLite and MySQL. Software: Python 3.13, scikit-learn 1.9, XGBoost 3.4, PyTorch 2.15,

(a) Executive dashboard (b) Network explorer (c) Disruption simulator (d) Gemini assistant

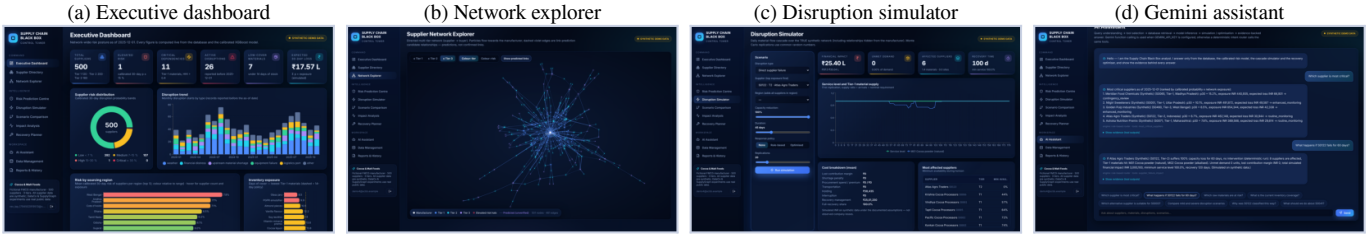


Fig. 5. Deployed application (React control tower, Flask API, MySQL/SQLite, Gemini tool-calling assistant; 35 automated tests pass, all pages browser-tested): (a) executive dashboard, (b) network explorer with predicted links, (c) disruption simulator, (d) evidence-backed assistant.

PyG 2.8, PuLP 2.9, React 18, Flask 3. **AI use declaration.** Claude Code (Anthropic) was used as a programming assistant for code scaffolding, debugging and test generation, and for language editing and checking of bibliographic details. Google Gemini is used only at run time inside the application's assistant, where it may phrase answers solely from tool outputs. All research design choices, experiments, results and interpretations were reviewed and verified by the authors, who take full responsibility for the content of this paper.

VII. Conclusion and Future Work

Calibrated prediction alone rarely justifies supplier-level action when buffers are healthy; the decisive value lies in exposing hidden dependencies and optimising recovery for severe shocks, where simulated impact fell by 75 %. Future work: validate on real supplier maps (bills of lading, UN Comtrade cocoa flows), learn risk-aware rolling-horizon policies that avoid over-hedging, and feed verified links back into training.

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